

Comparative Evaluation of Support Vector Machines and Baseline Classifiers on CIFAR-10 Subset

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Abstract—This project investigates the performance of Support Vector Machines (SVMs) and baseline classifiers on a subset of the CIFAR-10 dataset. The study applies Principal Component Analysis (PCA) to reduce the dimensionality of the input space while retaining over 90% of the variance, resulting in a compact representation of 91 features from the original 3072. Several models were implemented and evaluated, including k-Nearest Neighbors (1-NN, 3-NN), Nearest Class Centroid (NCC), a custom Linear SVM trained with stochastic gradient descent, scikit-learn SVMs with linear, polynomial, and radial basis function (RBF) kernels, and a Multi-Layer Perceptron (MLP) with hinge loss. Each classifier was assessed in terms of accuracy, training time, and testing time. Experimental results show that the RBF SVM achieved the highest accuracy (65.1%), outperforming both linear and polynomial kernels as well as baseline methods. Confusion matrices and visual examples of correct and incorrect classifications highlight the strengths and weaknesses of each approach. The findings confirm the effectiveness of kernelized SVMs for image classification tasks, while also illustrating the limitations of simpler baselines and shallow neural architectures.

I. INTRODUCTION

Image classification is a fundamental task in computer vision, with applications ranging from autonomous driving and medical imaging to content moderation and visual search. The CIFAR-10 dataset, consisting of 60,000 color images across 10 object categories, has become a widely used benchmark for evaluating classification algorithms due to its diversity and moderate complexity. Each image is relatively small (32×32 pixels), yet the dataset remains challenging because of intra-class variability and inter-class similarity.

Support Vector Machines (SVMs) are among the most powerful classical machine learning methods for classification. By maximizing the margin between decision boundaries, SVMs can achieve strong generalization performance, especially when combined with kernel functions that allow nonlinear separation of data. However, SVMs are computationally intensive on high-dimensional datasets,

which motivates the use of dimensionality reduction techniques such as Principal Component Analysis (PCA). PCA projects the data into a lower-dimensional subspace while preserving most of the variance, thereby reducing training time and mitigating overfitting.

This project focuses on evaluating the performance of SVMs on a subset of CIFAR-10, restricted to five classes for tractability. Several variants are considered: a custom linear SVM implemented with stochastic gradient descent, scikit-learn SVMs with linear, polynomial, and radial basis function (RBF) kernels, and baseline classifiers including k-Nearest Neighbors (k-NN), Nearest Class Centroid (NCC), and a Multi-Layer Perceptron (MLP) trained with hinge loss. Each model is assessed in terms of accuracy, training time, and testing time, with confusion matrices and sample predictions used to illustrate strengths and weaknesses.

II. DATASET AND PREPROCESSING

The experiments were conducted on a subset of the CIFAR-10 dataset, which consists of 60,000 color images of size 32×32 pixels across ten object categories. To reduce complexity and focus the analysis, five classes were selected: *airplane*, *automobile*, *bird*, *cat*, and *dog*. Each class provides sufficient variability to challenge the classifiers while keeping the computational requirements manageable.

A. Data Preparation

- **Loading the dataset:** The CIFAR-10 subset was loaded using the torchvision library. Training and test sets were separated according to the standard split (50,000 training images, 10,000 test images).
- **Class selection:** Only the five chosen categories were retained, and labels were remapped to values from 0 to 4.
- **Normalization:** Pixel values were scaled to the range [0,1] to ensure consistent input for all models.
- **Flattening:** Each image (32×32×3) was reshaped into a 3072-dimensional vector to serve as input features for classical machine learning algorithms.

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- **Raw test images:** A copy of the original test images was preserved (`x_test_raw`) for visualization of correct and incorrect predictions.

B. Dimensionality Reduction with PCA

High-dimensional input vectors can lead to long training times and risk of overfitting. To address this, Principal Component Analysis (PCA) was applied:

- **Variance retention:** PCA was configured to retain 90% of the variance in the data.
- **Reduced dimensionality:** The original 3072 features were reduced to 91 principal components.
- **Impact:** This transformation significantly reduced computational cost while preserving most of the discriminative information in the dataset.

By combining normalization, flattening, and PCA, the dataset was transformed into a compact and efficient representation suitable for training multiple classifiers. This preprocessing pipeline ensured that each model could be evaluated fairly under consistent conditions.

III. CLASSIFIERS AND METHODOLOGY

To evaluate the effectiveness of Support Vector Machines (SVMs) relative to baseline classifiers, several models were implemented and tested under consistent experimental conditions. Each classifier was trained on the PCA-transformed training set and evaluated on the PCA-transformed test set. Performance was measured in terms of accuracy, training time, and testing time, with confusion matrices and sample predictions used for qualitative analysis.

A. *k*-Nearest Neighbors (*k*-NN)

The *k*-NN algorithm classifies a test sample based on the majority label among its *k* nearest neighbors in the training set.

- **Variants:** 1-NN and 3-NN were tested.
- **Rationale:** Provides a simple non-parametric baseline.
- **Limitations:** Sensitive to noise and computationally expensive at test time.

B. Nearest Class Centroid (NCC)

NCC assigns each test sample to the class whose centroid (mean vector) is closest in feature space.

- **Rationale:** Extremely fast at test time and provides a simple linear baseline.
- **Limitations:** Assumes classes are well represented by their mean, which is often unrealistic for complex image data.

C. Custom Linear SVM (One-vs-Rest)

A custom implementation of a linear SVM was developed using stochastic gradient descent (SGD) and hinge loss.

- **Approach:** One-vs-Rest strategy was used to extend binary SVMs to multi-class classification.
- **Rationale:** Demonstrates the mechanics of SVM training without relying on external libraries.
- **Limitations:** Training stability and convergence depend heavily on learning rate and number of epochs.

D. scikit-learn SVMs

Three kernelized SVMs were trained using scikit-learn's implementation:

- **Linear SVM:** Uses a linear kernel, serving as a baseline for comparison.
- **Polynomial SVM:** Employs a polynomial kernel of degree 3, enabling nonlinear decision boundaries.
- **RBF SVM:** Uses the radial basis function kernel, which maps data into a high-dimensional space and is well suited for complex, nonlinear classification tasks.
- **Rationale:** Kernelized SVMs are expected to outperform linear models by capturing nonlinear relationships in the data.

E. Multi-Layer Perceptron (MLP) with Hinge Loss

A shallow neural network was implemented in NumPy to evaluate the performance of a non-kernelized nonlinear classifier.

- **Architecture:** One hidden layer with 128 units and ReLU activation.
- **Loss function:** Hinge loss, consistent with SVM training objectives.
- **Training:** Mini-batch SGD with learning rate 0.001, batch size 128, and 10 epochs.
- **Rationale:** Provides a neural baseline for comparison with SVMs.
- **Limitations:** Shallow architecture and limited training epochs restrict performance relative to deeper networks.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

All classifiers were trained and evaluated on the PCA-transformed CIFAR-10 subset (five classes: airplane, automobile, bird, cat, dog). PCA reduced the dimensionality from 3072 to 91 while retaining 90% of the variance. Training

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and testing were conducted on the same hardware under consistent conditions to ensure fair comparison. Accuracy, training time, and testing time were recorded for each model.

B. Quantitative Results

The table below summarizes the performance of all classifiers:

Model	Accuracy	Train Time (s)	Test Time (s)
1-NN	0.5132	0.0015	1.5090
3-NN	0.5028	0.0011	0.1383
NCC	0.3908	0.0000	0.0049
Custom Linear SVM	0.4652	—	—
Linear SVM	0.5350	1.4922	0.0018
RBF SVM	0.6514	20.4855	6.4829
Polynomial SVM	0.5636	26.4280	2.4142
MLP (hinge)	0.5108	—	—

The RBF SVM achieved the highest accuracy (65.1%), outperforming both linear and polynomial kernels as well as baseline methods. Polynomial SVM showed moderate improvement over linear SVM, while the custom linear SVM and MLP performed comparably to simple baselines.

C. Confusion Matrices

Confusion matrices were generated to visualize classification errors.

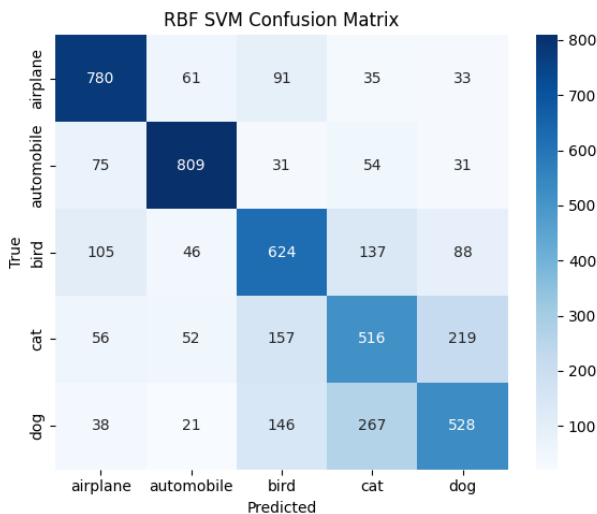


Figure 1. Confusion Matrix for RBF SVM

The RBF kernel achieved the highest accuracy, with strong separation between airplane and automobile classes. However, frequent misclassifications occurred between cats and dogs, reflecting their visual similarity.

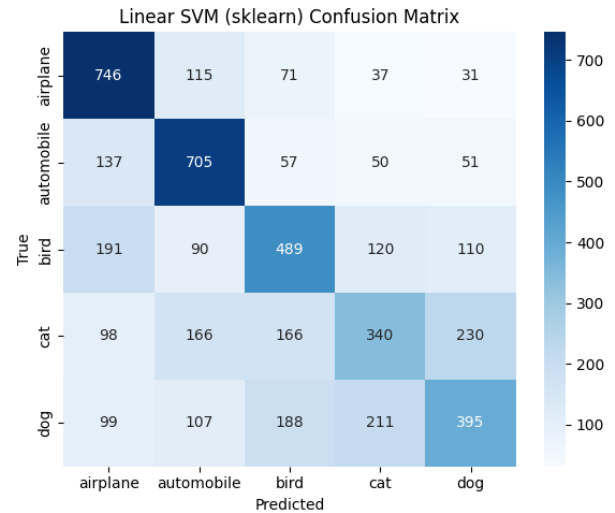


Figure 2. Confusion Matrix for Linear SV

The linear kernel showed weaker separation, with errors distributed across multiple categories. This highlights the limitations of linear decision boundaries in complex image data.

D. Qualitative Results

To further illustrate model behavior, sample predictions were visualized:

- **Correct predictions:** Images where the classifier successfully identified the true class.
- **Incorrect predictions:** Images where the classifier misclassified, often due to visual similarity (e.g., cats vs. dogs, birds vs. airplanes).



Figure 3. Correct Predictions (RBF SVM)

Representative test images correctly classified by the RBF SVM. These examples illustrate the model's ability to capture discriminative features across diverse categories.

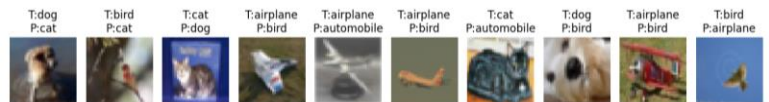


Figure 4. Incorrect Predictions (RBF SVM)

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Representative test images misclassified by the RBF SVM. Most errors occurred between visually similar categories (e.g., cats vs. dogs), underscoring the challenge of fine-grained classification in CIFAR-10.

V. DISCUSSION

The experimental results reveal clear differences in performance across the evaluated classifiers. Baseline methods such as k-NN and NCC provided useful points of comparison but demonstrated inherent limitations. While 1-NN achieved moderate accuracy (51.3%), its reliance on individual neighbors made it sensitive to noise and computationally expensive at test time. NCC, although extremely fast, performed poorly (39.1%) because class centroids failed to capture the variability within each category. These findings confirm that simple distance-based methods are insufficient for complex image classification tasks.

Linear models, including the custom Linear SVM and scikit-learn's Linear SVM, improved upon the baselines by leveraging margin maximization. However, their accuracy remained limited (46.5% and 53.5%, respectively), reflecting the inability of linear decision boundaries to separate classes with nonlinear distributions. Polynomial SVM offered modest improvement (56.4%), but its performance was constrained by the fixed polynomial degree and increased computational cost.

The RBF SVM achieved the highest accuracy (65.1%), demonstrating the effectiveness of kernel methods in capturing nonlinear relationships. The RBF kernel maps data into a high-dimensional space, enabling the classifier to construct flexible decision boundaries that better separate visually similar categories. This advantage was particularly evident in distinguishing airplanes from birds and automobiles, though challenges persisted in separating cats and dogs due to their high visual similarity.

The Multi-Layer Perceptron (MLP) with hinge loss provided a neural baseline but did not outperform kernelized SVMs. Its accuracy (51.1%) was comparable to k-NN, reflecting the limitations of shallow architectures trained for a small number of epochs. Deeper networks, longer training schedules, or data augmentation would likely improve performance, but such extensions were beyond the scope of this study.

VI. CONCLUSIONS

This study evaluated the performance of Support Vector Machines (SVMs) and baseline classifiers on a subset of the CIFAR-10 dataset, reduced to five classes and transformed via Principal Component Analysis (PCA). The experiments demonstrated that kernelized SVMs, particularly the RBF variant, significantly outperformed linear models and simple baselines. By mapping data into a higher-dimensional space, the RBF kernel enabled flexible decision boundaries that

captured complex nonlinear relationships, achieving the highest accuracy of 65.1%.

Baseline methods such as k-NN and NCC provided useful benchmarks but were limited by sensitivity to noise and oversimplified assumptions about class distributions. Linear SVMs improved upon these baselines but remained constrained by their inability to separate nonlinear data. Polynomial SVMs offered moderate gains but at the cost of increased computational complexity. The Multi-Layer Perceptron (MLP) with hinge loss highlighted the potential of neural approaches but underperformed due to its shallow architecture and limited training schedule.

Three main conclusions can be drawn:

- **Kernelized SVMs are the most effective classical classifiers** for this task, outperforming both linear models and shallow neural networks.
- **Dimensionality reduction via PCA was essential** for enabling efficient training while preserving most of the dataset's variance.
- **Shallow neural architectures are insufficient** for complex image classification compared to well-tuned kernel methods, though deeper networks could potentially close the gap.

Future work could explore deeper neural networks with longer training schedules, data augmentation to improve generalization, and hyperparameter optimization for SVMs. Extending the analysis to all ten CIFAR-10 classes or to more complex datasets would further validate the findings. Overall, the results confirm the strength of kernelized SVMs as a robust choice for image classification in settings where computational resources are limited and deep learning is impractical.

REFERENCES

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3. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

APPENDIX NOTE: USE OF CHATGPT

During the development of this project, ChatGPT was used as a supportive tool to:

- Help structure the project workflow and organize the report sections.

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- Provide guidance in debugging code and improving visualization functions.
- Suggest clear explanations of algorithms and experimental methodology.